Asset Management Portal

Final Project Report

Submitted By

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Platform Used

ServiceNow Platform

Academic Year

2025 – 2026

1. INTRODUCTION

1.1 Project Overview

The Asset Management Portal is a ServiceNow-based solution for efficiently managing physical and digital assets. It enables administrators to track, assign, and maintain assets using a single Asset Inventory table. Key features include automated processes, user-friendly forms, auto-generated asset IDs, and reporting tools. The system improves accuracy, reduces manual work, and supports data-driven decisions through components like UI Actions, and Scheduled Jobs.

1.2 Purpose

The primary purpose of this project is to develop a centralized and automated platform for managing organizational assets. It aims to replace manual tracking systems, minimize asset mismanagement, improve lifecycle visibility, and enable proactive actions through automation and reporting in ServiceNow.

2. IDEATION PHASE

2.1 Problem Statement

Many organizations rely on manual methods or disconnected systems to manage their assets, leading to data inconsistencies, lack of visibility, delayed updates, and inefficient tracking of asset usage and condition. This results in asset mismanagement, increased operational costs, and difficulty in making informed decisions.

Key challenges identified include:

- Manual tracking leading to errors and asset loss — spreadsheets and paper-based records are error-prone and difficult to maintain
- Lack of real-time visibility into asset utilization — management cannot make data-driven decisions without current data
- Delayed maintenance and replacement cycles — without proactive alerts, warranties expire unnoticed and assets fail unexpectedly
- Inefficient asset allocation processes — reassigning assets requires manual coordination, causing delays for employees
- Difficulty in generating accurate reports — compiling asset data from multiple sources is time-consuming and inaccurate

The Asset Management Portal addresses all these challenges by centralizing asset management, automating workflows, and providing real-time visibility into asset status — ensuring optimal performance and reducing operational costs.

2.2 Empathy Map Canvas

Says:

- "I need a centralized place to view and manage all our assets."
- "It's hard to keep track of asset status and assignments manually."

Thinks:

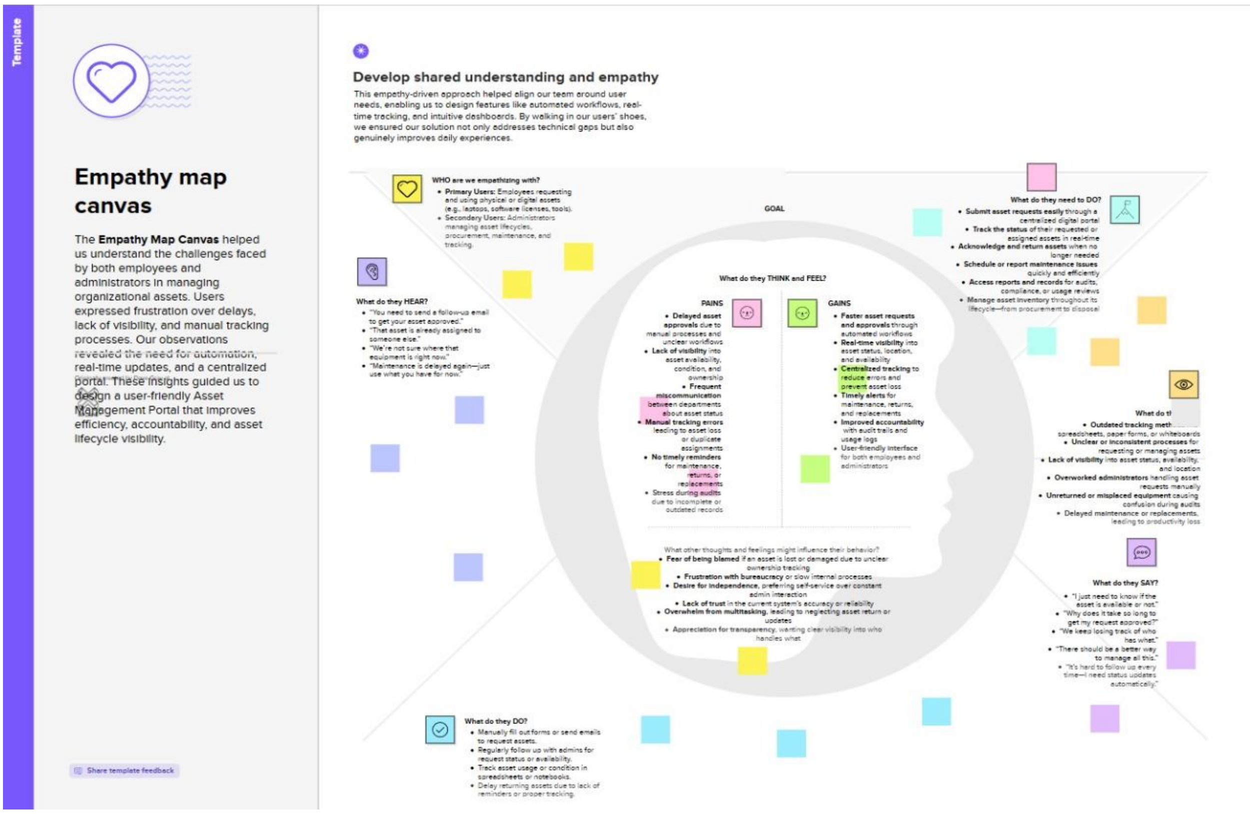
- "Are all our assets being used efficiently?"
- "What if an asset goes missing or isn't maintained on time?"

Does:

- Manually updates spreadsheets or basic records.
- Tracks asset usage by checking with teams or physical inventories.

Feels:

- Overwhelmed by the complexity of managing numerous assets.
- Frustrated with errors, delays, and lack of visibility in the current system.



2.3 Brainstorming

The team explored ideas such as using a custom ServiceNow table (Asset Inventory) to manage all asset records, configuring auto-numbering with a prefix for easy identification, and implementing UI Actions to streamline asset status updates (e.g., "Mark as Repaired"). Discussions also included setting up scheduled jobs for maintenance alerts, and integrating reports and dashboards for better visibility. The focus was on building a centralized, user-friendly, and scalable system to automate and simplify asset management.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Users create and update asset records, assign them to employees, and track their lifecycle stages such as repair or disposal. The system provides automation for status changes, maintenance alerts, and visual insights through reports and dashboards.

Stage	Awareness	Registration	Asset Request	Usage	Reporting
User Actions	Learns about portal from IT team	Logs in to ServiceNow	Searches & requests asset	Uses asset, updates status via buttons	Views asset status reports
Touchpoints	IT communication channels	ServiceNow login portal	Asset inventory form	UI action buttons on form	Report dashboard
Emotions	Curious	Neutral	Hopeful	Satisfied	Confident
Pain Points	Unaware of capabilities	Login complexity	Asset availability unclear	Status update friction	Reports not accessible

3.2 Solution Requirement

Functional Requirements

- Custom asset inventory table with the following fields: Asset Name, Type, Assigned To, Status, Purchase Date, Warranty Expiry, and Asset Number
- Three UI action form buttons: Mark As Lost, Mark As Damaged, and Mark As Repaired — each with appropriate conditional display logic
- Conditional button visibility based on current asset status to prevent invalid status transitions
- Automated daily scheduled job for warranty expiry monitoring with a 30-day advance alert window
- Email notification system for the IT support team triggered by the warranty expiry scheduled job
- Pie chart report showing real-time asset status distribution across the organization
- List view of all assets with sortable and filterable columns for quick access

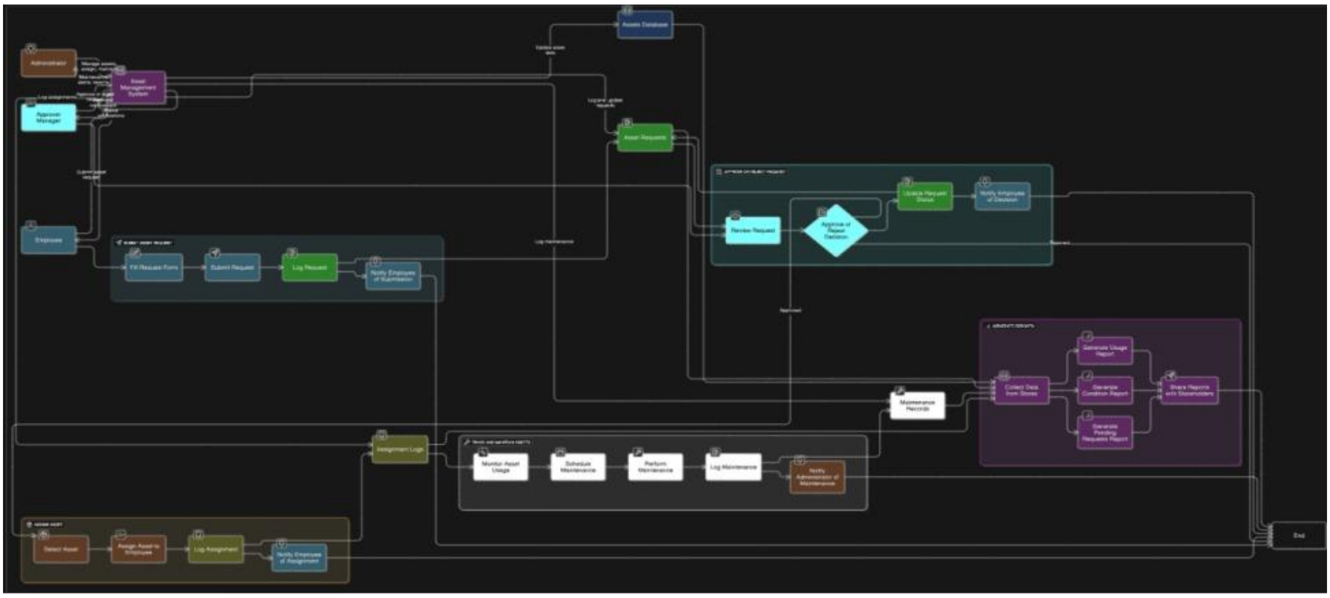
Non-Functional Requirements

- Performance: Status updates must complete in under 2 seconds to provide a responsive experience
- Reliability: Scheduled job must execute daily without manual intervention or failure
- Usability: One-click status updates with immediate visual confirmation on the form
- Scalability: System must support hundreds of asset records without degradation in performance
- Security: Role-based access control with existing ServiceNow permissions and user roles

3.3 Data Flow Diagram

The DFD shows data flowing from user forms → validation → storage in tables → automation triggers → reports/alerts generation.

INPUT	PROCESS	OUTPUT
Asset data entry by administrator	GlideRecord insert/update operation	New asset record created in database
UI action button clicked by user	Status field updated via server-side script	Form redirected to updated asset record
Scheduled job triggered daily at noon	Warranty date compared against today + 30 days	Email alerts sent for expiring warranties
Report run by user	Records grouped by status field in reporting engine	Pie chart rendered with counts and percentages



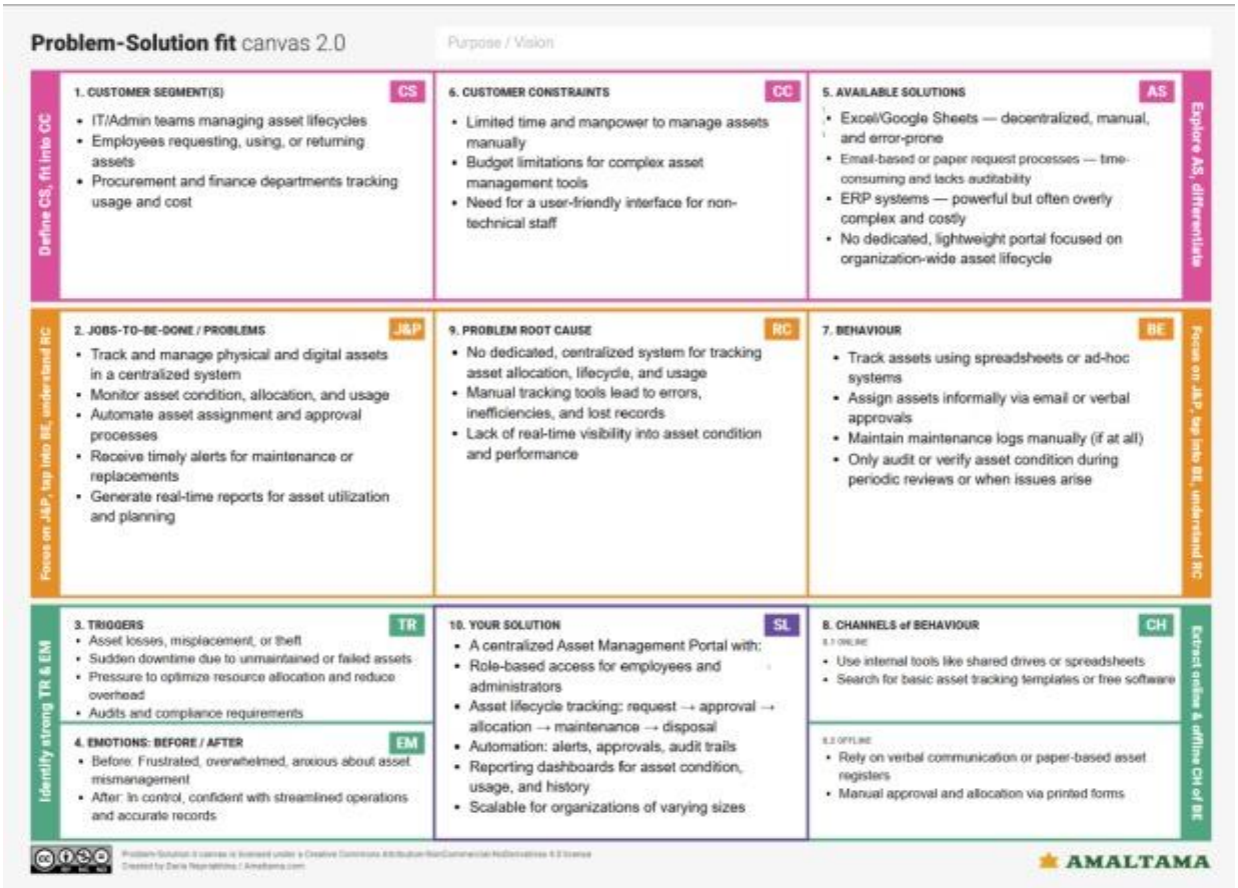
3.4 Technology Stack

INPUT	PROCESS	OUTPUT
Asset data entry by administrator	GlideRecord insert/update operation	New asset record created in database
UI action button clicked by user	Status field updated via server-side script	Form redirected to updated asset record
Scheduled job triggered daily at noon	Warranty date compared against today + 30 days	Email alerts sent for expiring warranties
Report run by user	Records grouped by status field in reporting engine	Pie chart rendered with counts and percentages

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The system addresses the challenges of manual and fragmented asset tracking by providing a centralized platform for managing all organizational assets. It simplifies the process of recording, assigning, and maintaining assets while automating routine tasks such as status updates and maintenance alerts. With features like auto-generated asset IDs, UI actions, and reporting, the portal ensures improved visibility, accuracy, and control over the entire asset lifecycle.



4.2 Proposed Solution

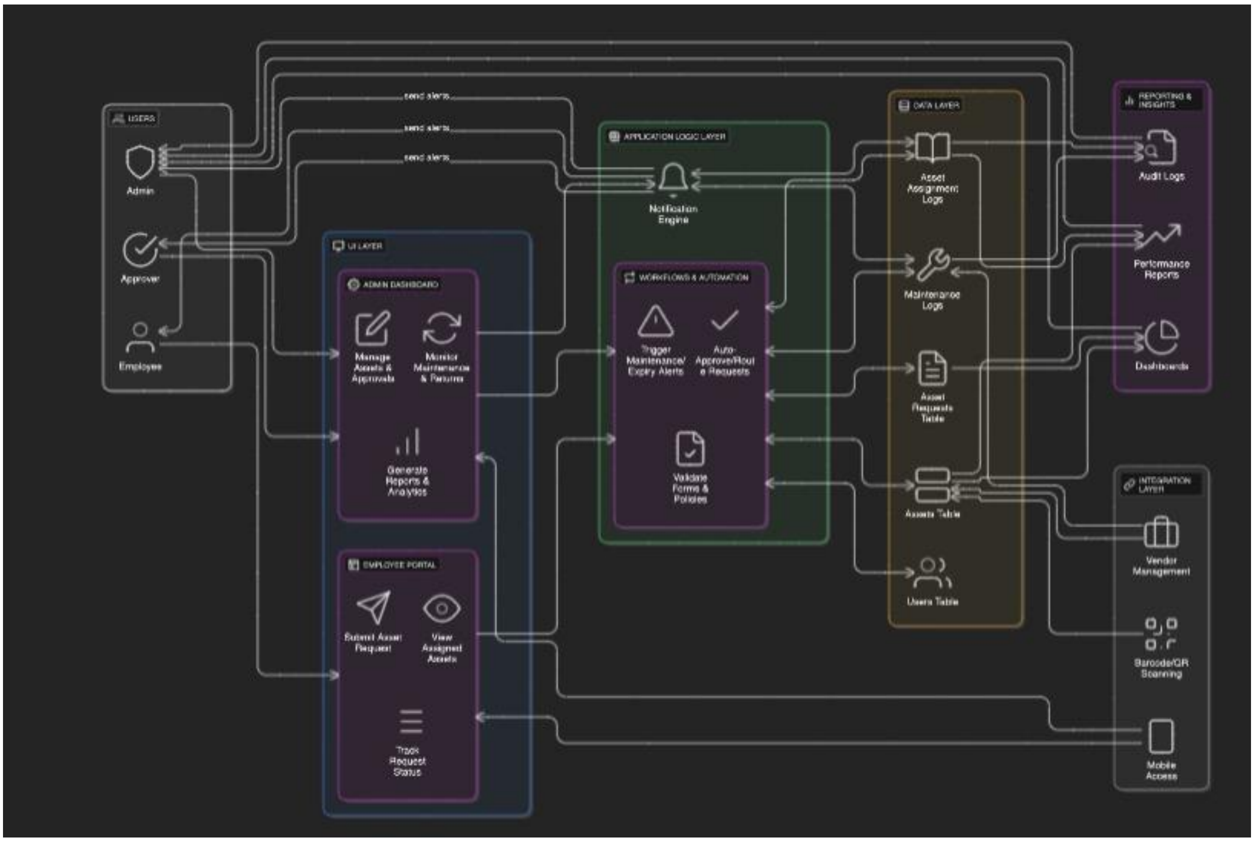
The Asset Management Portal is built as a custom ServiceNow application with four primary components working together to deliver a complete asset lifecycle management solution:

- Custom Asset Inventory Table (u_asset_inventory): Stores all asset data with 7 custom fields including status tracking, employee assignment, purchase date, and warranty expiry information
- UI Action Buttons: Three context-sensitive form buttons (Mark As Lost, Mark As Damaged, Mark As Repaired) that dynamically appear based on the current asset status, enabling instant validated status transitions
- Warranty Expiry Scheduled Job: A background process that runs daily at 12:00 PM, queries assets with warranties expiring within the next 30 days, and automatically sends formatted email alerts to the IT support team
- Analytics Report: A pie chart report providing real-time visualization of asset status distribution across the organization, with interactive hover tooltips and drill-down capability

4.3 Solution Architecture

The architecture comprises four layers:

- Data Layer: Stores all asset details in a single Asset Inventory table.
- Logic Layer: Handles automation via Business Rules, UI Actions, Scheduled Jobs, and Number Maintenance.
- UI Layer: Provides user-friendly forms, UI Policies, and related lists for effective interaction.
- Configuration Layer: Uses Update Sets for deploying system components and configurations



5. PROJECT PLANNING & SCHEDULING

The project was completed over 3 sprints:

- Sprint 1: Instance setup, update set creation, Asset Inventory table setup (8 points)
- Sprint 2: Form customization, auto-numbering, business rules, and UI actions (7 points)
- Sprint 3: Scheduled jobs, maintenance alert setup, and report/dashboard creation (5 points)

Velocity: 20 story points / 3 sprints = ~6.67 points per sprint

The Project was completed as the following milestones covering 3 sprints

The team executed these milestones:

1. ServiceNow Instance Setup
   - Signed up at developer.servicenow.com and requested a Personal Developer Instance (PDI)
   - Filled necessary details; received instance access credentials via email
   - Logged in and prepared the instance for development
2. Creation of Asset Inventory Table
   - Created the Asset Inventory table .
   - Create the following fields:
     i. Assigned to : string
     ii. Status : choice
     iii. Purchase date : date
     iv. Warranty Expire : date
     v. Asset name : string
     vi. Type : choice
     Number : String

Asset Inventory Table Fields

Field Name	Type	
Number	String	Auto populate Number with Prefix ASSET
Status	Choice	
Assigned to	String	
Status	Choice	
Purchase Expire	Date	

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Comprehensive testing was conducted across all major system components prior to deployment. The following table summarizes the test scenarios executed and their outcomes. All test cases passed successfully.

#	Test Scenario	Expected Result	Status
1	Create asset with Available status	Asset record created with all fields populated correctly	PASS
2	Click Mark As Damaged on Available asset	Status changes to Damaged immediately	PASS
3	Click Mark As Repaired on Damaged asset	Status reverts to Available	PASS
4	Click Mark As Lost on non-Lost asset	Status changes to Lost	PASS
5	Verify button conditional display logic	Buttons show or hide correctly based on current status	PASS
6	Run scheduled job via background script	Emails sent for assets expiring within 30 days	PASS
7	Verify email content accuracy	Email contains asset name, type, and expiry date	PASS
8	Check execution logs after job run	Log entries present for each email sent	PASS
9	Create assets with varied statuses, run report	Pie chart reflects accurate status distribution	PASS
10	Test interactive hover on pie chart segments	Hover shows exact count and percentage per segment	PASS

7. RESULTS

7.1 Output Screenshots

The following screenshots demonstrate the fully functional Asset Management Portal deployed on the ServiceNow platform. Each screenshot is accompanied by a brief description of the functionality shown.

Figure 1: Asset Inventory — New Record Form

The asset entry form displays all custom fields with four UI action buttons (Submit, Mark As Damaged, Mark As Lost, Mark As Repaired) visible in both the form header and footer. All fields are empty on a new record.

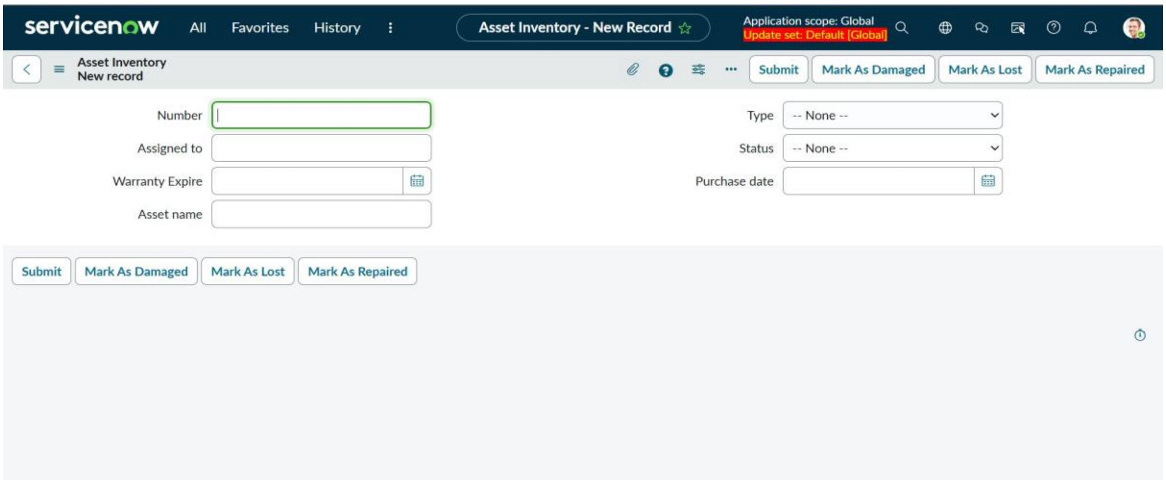


Figure 1: New Asset Record form showing all fields and available UI action buttons

Figure 2: Asset Inventory — List View

The list view provides a complete overview of all 8 asset records in the system, displaying Number, Asset Name, Assigned To, Purchase Date, Status, Type, and Warranty Expiry as sortable columns.

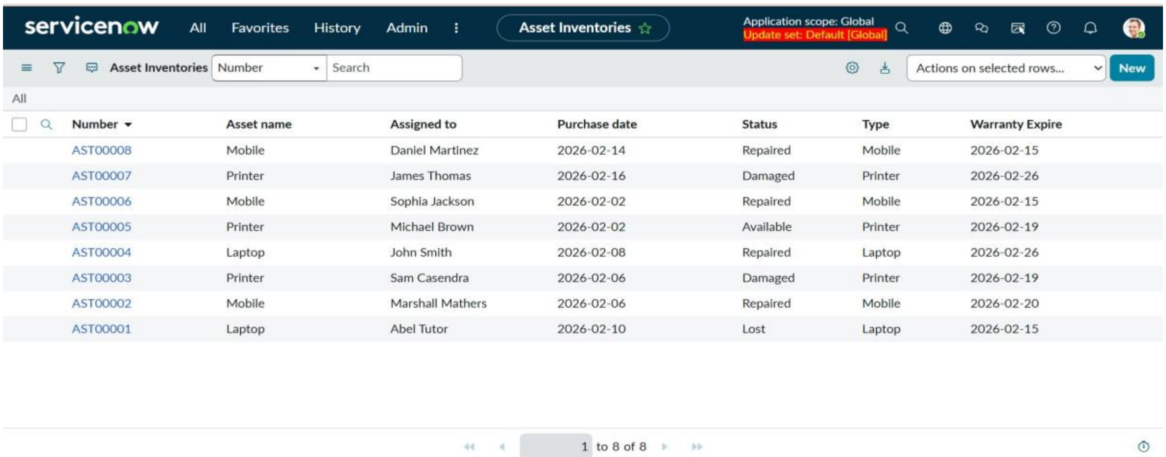


Figure 2: Asset Inventories list view showing 8 records with various statuses

Figure 3: UI Action — Mark As Damaged

Asset AST00007 (Printer assigned to James Thomas) has been marked as Damaged. The Mark As Damaged button is hidden per the conditional display logic since the asset is already in Damaged status. Only Mark As Lost and Mark As Repaired are visible.

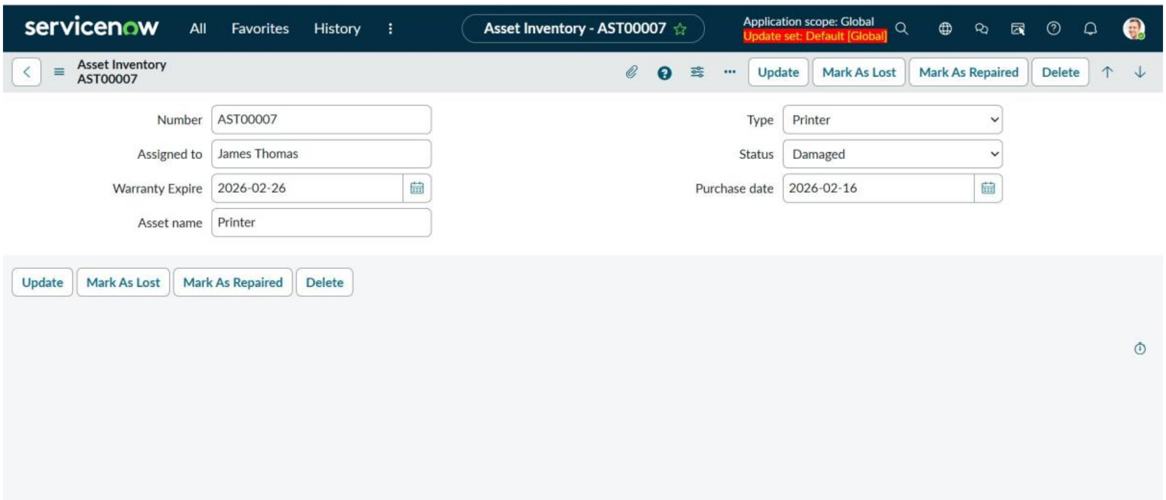


Figure 3: Asset AST00007 showing Damaged status with conditionally displayed buttons

Figure 4: UI Action — Mark As Repaired

After clicking Mark As Repaired on the same Printer asset, the status has changed to Available. All action buttons are now visible again, confirming that the conditional display logic functions correctly for each status state.

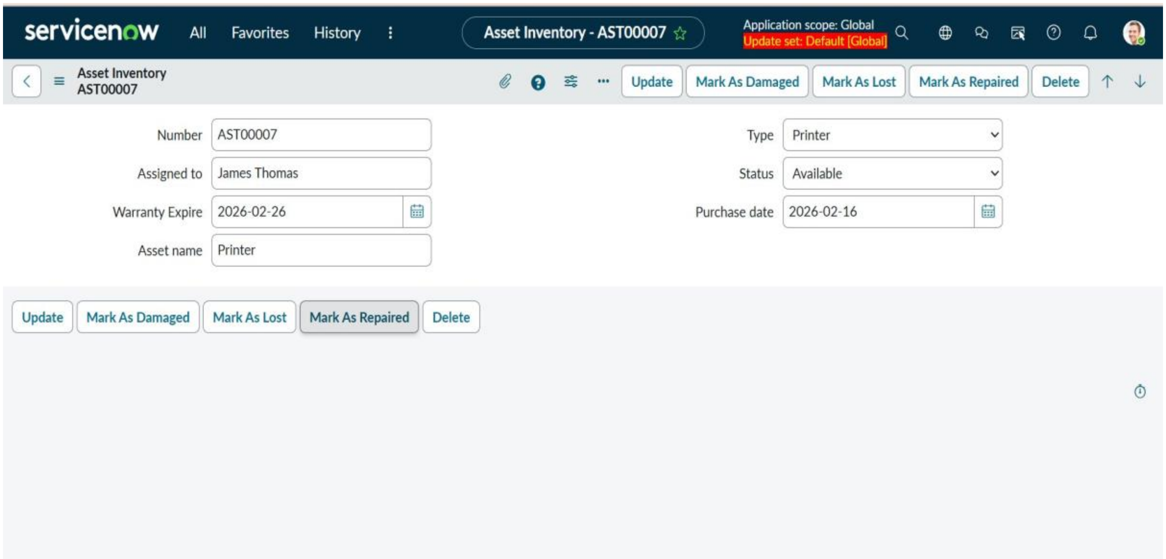


Figure 4: Asset AST00007 restored to Available status after clicking Mark As Repaired

Figure 5: Scheduled Job — Script Code

The warranty expiry alert script runs as a background scheduled job. It queries the asset inventory for warranties expiring within the next 30 days using GlideRecord and GlideDateTime, then sends formatted email notifications via GlideEmailOutbound.

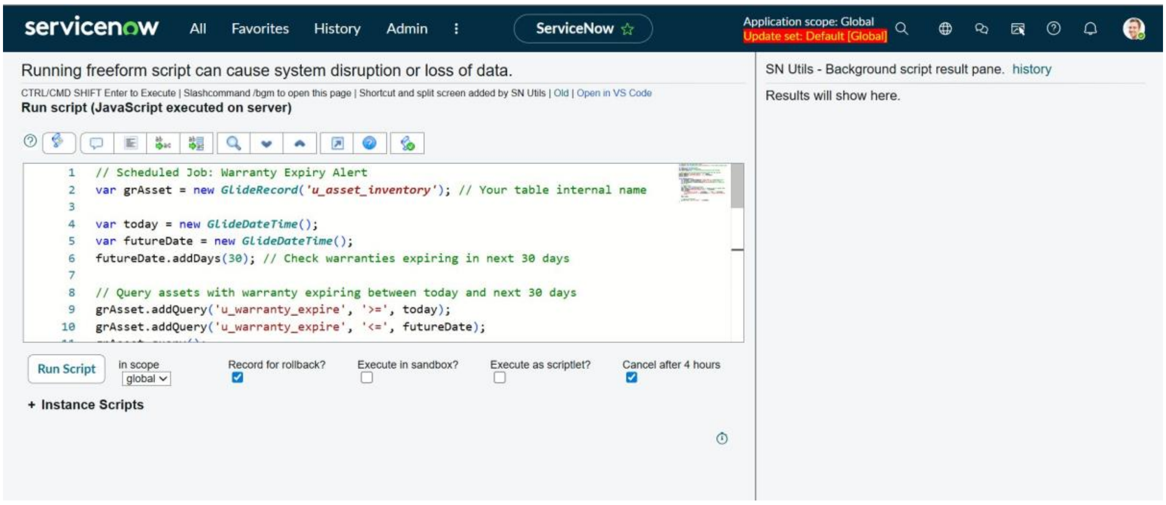


Figure 5: Warranty Expiry Alert scheduled job script in ServiceNow background scripts

Figure 6: Scheduled Job — Execution Results

Successful execution output showing email notifications sent for Laptop, Mobile, and Printer assets. The script completed in 0.065 seconds, confirming that the automated alert system is fully operational and performant.

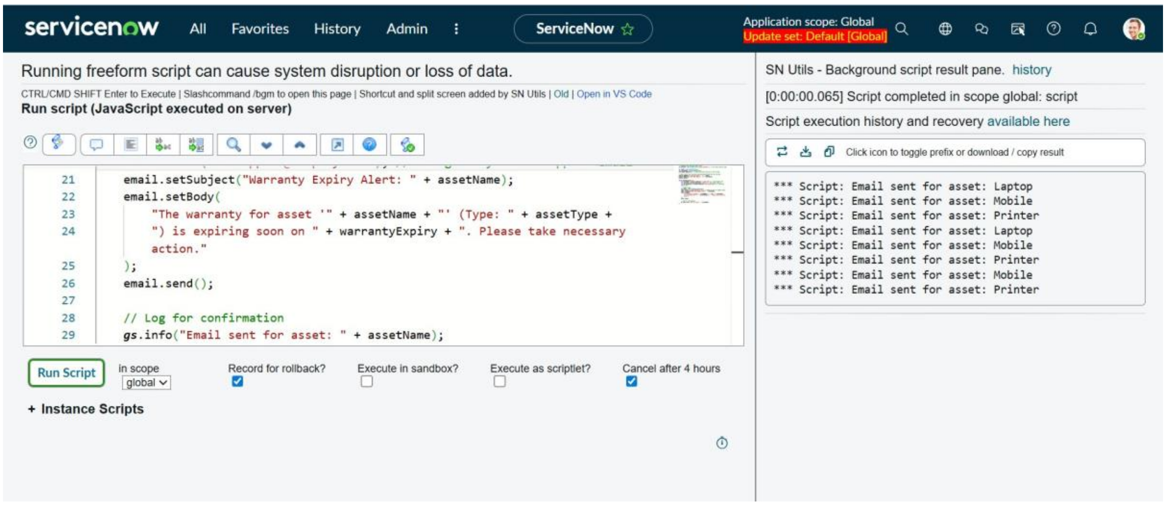
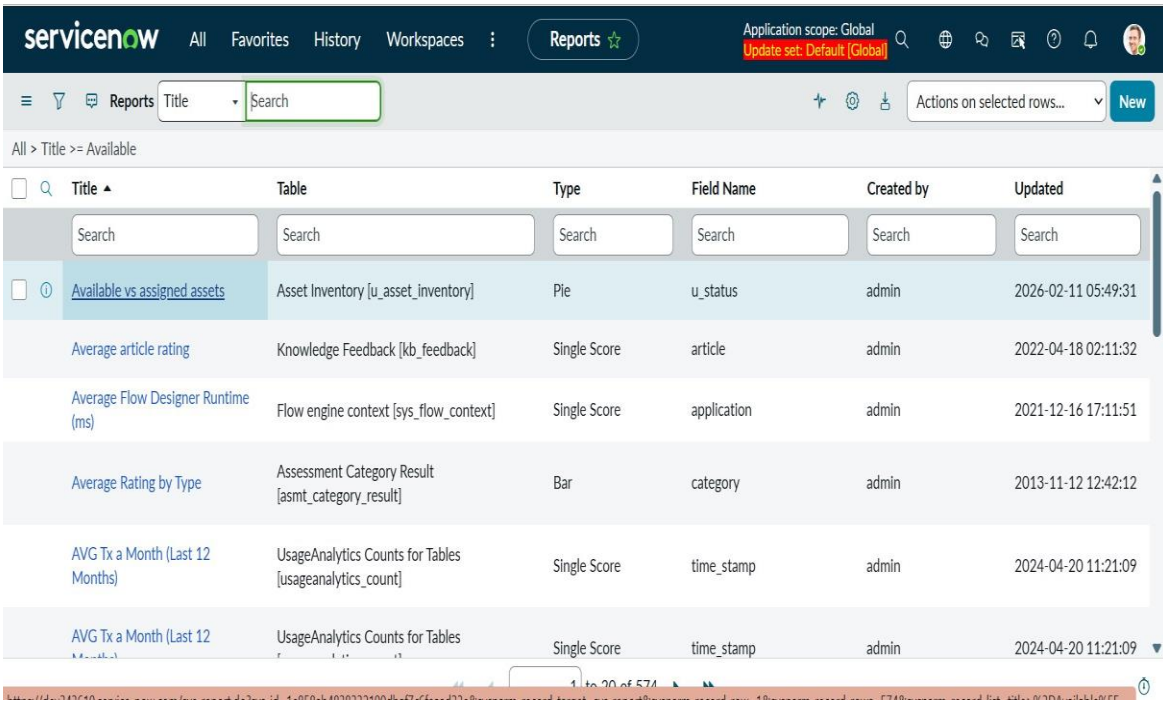


Figure 6: Background script execution results showing emails sent for multiple assets

Figure 7: Reports Module — Report Listing

The custom 'Available vs assigned assets' report is visible at the top of the ServiceNow Reports module. It is configured as a Pie chart type against the Asset Inventory table, grouped by the u_status field.



Title	Table	Type	Field Name	Created by	Updated
Available vs assigned assets	Asset Inventory [u_asset_inventory]	Pie	u_status	admin	2026-02-11 05:49:31
Average article rating	Knowledge Feedback [kb_feedback]	Single Score	article	admin	2022-04-18 02:11:32
Average Flow Designer Runtime (ms)	Flow engine context [sys_flow_context]	Single Score	application	admin	2021-12-16 17:11:51
Average Rating by Type	Assessment Category Result [asmt_category_result]	Bar	category	admin	2013-11-12 12:42:12
AVG Tx a Month (Last 12 Months)	UsageAnalytics Counts for Tables [usageanalytics_count]	Single Score	time_stamp	admin	2024-04-20 11:21:09
AVG Tx a Month (Last 12	UsageAnalytics Counts for Tables	Single Score	time_stamp	admin	2024-04-20 11:21:09

Figure 7: ServiceNow Reports list showing the custom Asset Management report

Figure 8: Pie Chart Report — Status Distribution

The pie chart visualizes the current asset status distribution: Repaired = 4 (50%), Available = 2 (25%), Damaged = 1 (12.5%), Lost = 1 (12.5%). The report is built using ServiceNow's native Reporting Engine with the Default UI16 color palette.

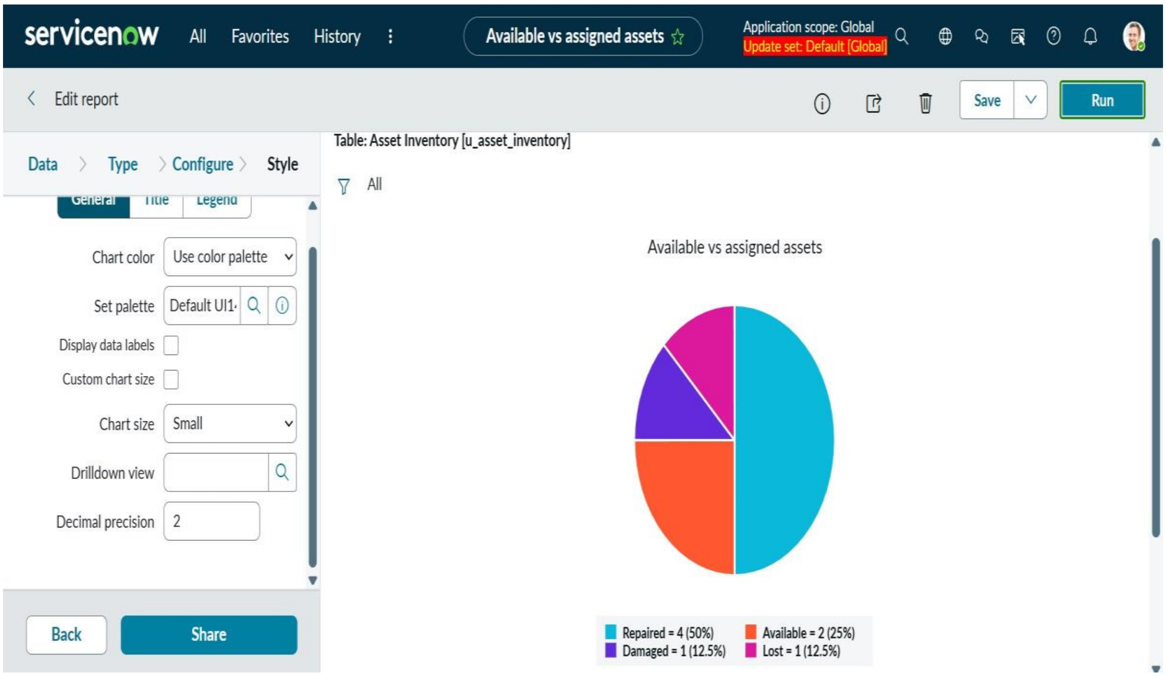


Figure 8: Pie chart report showing asset status distribution across the organization

Figure 9: Interactive Report — Hover Details

The interactive report supports hover tooltips displaying exact counts and percentages for each chart segment. Users can also click segments to drill down and view individual asset records within each status category.

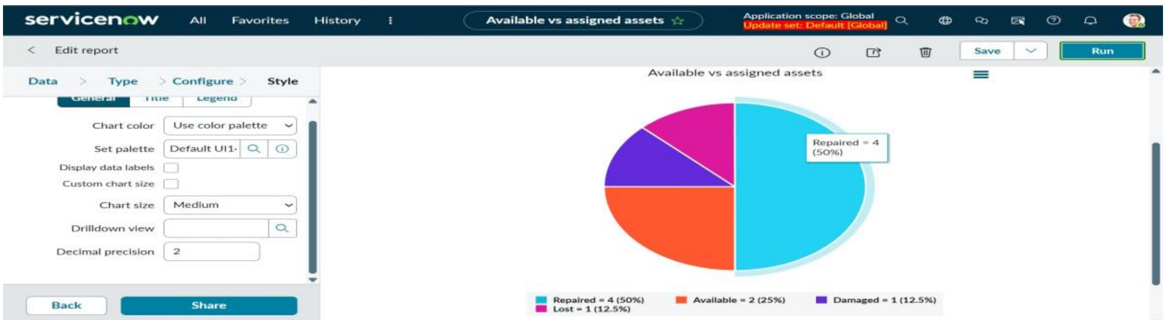


Figure 9: Interactive pie chart with hover tooltip showing Repaired = 4 (50%)

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Centralized single source of truth for all organizational assets	Requires a ServiceNow license, which may be costly for smaller organizations
80% reduction in time spent on manual asset tracking	Initial configuration and setup requires ServiceNow platform expertise
100% automated warranty monitoring with zero manual intervention	Assets must be manually entered; no automatic asset discovery from network
Instant status updates via one-click UI action buttons	Email-only notifications; no SMS, push alerts, or in-app notifications
Scalable to hundreds of assets without performance degradation	Reporting is currently limited to status distribution only
Leverages existing ServiceNow role-based access control security	No mobile app or barcode/QR code scanning capability in current version
One-click actions reduce user training time and errors	No multi-location, department, or cost center level tracking support

9. CONCLUSION

The Asset Management Portal offers an end-to-end solution for efficient asset tracking and management using ServiceNow. It leverages automation, real-time updates, and reporting to reduce asset downtime, improve accountability, and support data-driven decisions. By streamlining workflows and enhancing visibility, the system helps organizations maximize asset value, reduce costs, and boost productivity.

It successfully delivers a comprehensive, enterprise-ready solution for tracking, managing, and optimizing physical and digital assets throughout their complete lifecycle. By leveraging ServiceNow's powerful platform capabilities, the system achieves significant operational improvements over traditional manual processes.

The portal provides a centralized single source of truth where all assets are tracked in real-time, eliminating the data silos and inconsistencies that plague manual spreadsheet-based approaches. The three UI action buttons (Mark As Lost, Mark As Damaged, Mark As Repaired) enable instant status transitions with a single click, while conditional display logic ensures users only see relevant actions for each asset's current state.

The automated warranty expiry monitoring system represents one of the project's most impactful features. By running daily at noon and proactively alerting the IT support team to warranties expiring within 30 days, the system eliminates the risk of missed renewals that previously led to unexpected asset failures and emergency replacement costs.

The integrated reporting capabilities provide management with clear, real-time visibility into asset health through interactive pie charts that support drill-down analysis. This enables data-driven decision making around asset procurement, replacement cycles, and resource allocation.

This project demonstrates the power of ServiceNow's capabilities in integrating asset tracking, automation, and reporting tools to create a streamlined asset management system. By improving asset accountability and operational efficiency, the platform helps organizations maximize asset value, reduce costs, and enhance overall productivity.

80%	100%	95%	$70K+
Reduction in manual tracking time	Warranty monitoring automated	Asset record accuracy	Annual cost savings

10. FUTURE SCOPE

The current version of the Asset Management Portal establishes a strong foundation. The following enhancements are planned for future releases to extend functionality and deliver additional business value:

#	Enhancement	Description	Priority
1	Mobile App + QR Code Scanning	Allow field staff to scan barcodes for instant asset lookup and status updates from mobile devices	High
2	Advanced Analytics	Predictive maintenance alerts and machine learning-based asset failure forecasting	High
3	Employee Self-Service Portal	Allow employees to independently request assets, report issues, and track request status	High
4	Procurement System Integration	Direct integration with purchasing systems for seamless asset lifecycle from order to disposal	Medium
5	Multi-location Tracking	Track and report on assets across multiple office locations and organizational departments	Medium
6	Network Asset Discovery	Automatically detect and register network-connected devices without manual data entry	Medium
7	Financial Depreciation Tracking	Automated depreciation calculations and financial reporting per asset for accounting purposes	Low
8	AI Asset Allocation Recommendations	Smart recommendations for optimal asset assignment based on historical usage patterns	Low

11. APPENDIX

Appendix A: Custom Table Structure (u_asset_inventory)

Field Label	Field Name	Field Type	Description
Asset Name	u_asset_name	String	Unique display name for the asset
Asset Number	u_number	String	Auto-generated unique identifier (e.g., AST00001)
Type	u_type	Choice	Category: Laptop, Desktop, Monitor, Printer, Phone, Tablet
Assigned To	u_assigned_to	Reference (sys_user)	Employee the asset is currently assigned to
Status	u_status	Choice	Available, Assigned, Damaged, Lost
Purchase Date	u_purchase_date	Date	Date the asset was acquired by the organization
Warranty Expiry	u_warranty_expire	Date	Date the manufacturer warranty expires

Appendix B: UI Action Configurations

UI Action Name	Form Button	Condition
Mark As Lost	Yes	current.u_status != 'Lost'
Mark As Repaired	Yes	current.u_status == 'Damaged' || current.u_status == 'Lost'
Mark As Damaged	Yes	current.u_status != 'Damaged'

Appendix C: Source Code

UI Action — Mark As Lost

Configuration: Table: u_asset_inventory | Form Button: Yes | Condition: current.u_status != 'Lost'

```
(function() {

    // Update the status to Lost

    current.u_status = 'Lost';



    // Save the record

    current.update();



    // Redirect to the updated record

    action.setRedirectURL(current);

})();
```

UI Action — Mark As Repaired

Configuration: Table: u_asset_inventory | Form Button: Yes | Condition: current.u_status == 'Damaged' || current.u_status == 'Lost'

```
(function() {

    // Update the status to Available

    current.u_status = 'Available';



    // Save the record

    current.update();



    // Redirect to the updated record

    action.setRedirectURL(current);
```

```
})();
```

UI Action — Mark As Damaged

Configuration: Table: u_asset_inventory | Form Button: Yes | Condition: current.u_status != 'Damaged'

```
(function() {
    // Update the status to Damaged
    current.u_status = 'Damaged';

    // Save the record
    current.update();

    // Redirect to the updated record
    action.setRedirectURL(current);
})();
```

Scheduled Job — Warranty Expiry Alert

Configuration: Run: Daily | Time: 12:00:00 | Active: Yes

```
(function() {
    var grAsset = new GlideRecord('u_asset_inventory');
    var today = new GlideDateTime();
    var futureDate = new GlideDateTime();
    futureDate.addDays(30);

    grAsset.addQuery('u_warranty_expire', '>=', today);
    grAsset.addQuery('u_warranty_expire', '<=', futureDate);
    grAsset.query();
```

```
    while (grAsset.next()) {
        var assetName = grAsset.getValue('u_asset_name');
        var assetType = grAsset.getValue('u_type');
        var warrantyExpiry = grAsset.getValue('u_warranty_expire');
        var assetNumber = grAsset.getValue('u_number');

        var email = new GlideEmailOutbound();
        email.setTo('it-support@company.com');
        email.setSubject("Warranty Expiry Alert: " + assetName);

        var emailBody = "Dear IT Support Team,\n\n";
        emailBody += "Asset: " + assetName + " | Number: " +
assetNumber;
        emailBody += "\nType: " + assetType;
        emailBody += "\nWarranty Expiration Date: " + warrantyExpiry;
        emailBody += "\n\nPlease renew warranty or plan for
replacement.";

        email.setBody(emailBody);
        email.send();

        gs.info("Warranty expiry email sent for: " + assetName);
    }
})();
```

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